

Problem 1.1

Let μ be an outer measure on a set X and let $\{A_k\}_{k=1}^{\infty}$ be a sequence of subsets of X .

- (a) Prove that if $\mu[A_1] = 0$, then A_1 is μ -measurable.
- (b) Prove that A_1 is μ -measurable if and only if its complement A_1^c is μ -measurable.
- (c) Assume in the sequel that $\{A_k\}_{k=1}^{\infty}$ is a sequence of μ -measurable sets. Prove that if $A_1 \subseteq A_2 \subseteq \dots \subseteq A_k \subseteq \dots$, then $\mu[A_1] + \mu[A_2 \setminus A_1] + \dots + \mu[A_k \setminus A_{k-1}] = \mu[A_k]$.
- (d) Prove that if $A_1 \supseteq A_2 \supseteq \dots \supseteq A_k \supseteq \dots$ and $\mu[A_1] < +\infty$ then $\lim_{k \rightarrow +\infty} \mu[A_k] = \mu[\bigcap_{k=1}^{\infty} A_k]$. Give a counterexample to this statement when $\mu[A_1] = +\infty$.
- (e) Assume that $B \subseteq X$. Show that if A_1 is μ -measurable then A_1 is $\mu|_B$ -measurable.

Solution. (a): Suppose $\mu[A_1] = 0$. To show that A_1 is μ -measurable, it suffices to show that $\mu[B] \geq \mu[B - A_1] + \mu[A_1 \cap B]$ for all $B \subseteq X$. Fix $B \subseteq X$. Note that $B - A_1 \subseteq B$ and $A_1 \cap B \subseteq A_1$ hence $\mu[B - A_1] \leq \mu[B]$ and $\mu[A_1 \cap B] = 0$. Hence $\mu[B] \geq \mu[B - A_1] = \mu[B - A_1] + \mu[A_1 \cap B]$.

(b): Suppose A is μ -measurable, thus $\mu[B] = \mu[B - A] + \mu[A \cap B]$. We want to show that $\mu[B] \geq \mu[B - A^c] + \mu[A^c \cap B]$ for all $B \subseteq X$.

$$\begin{aligned} B - A^c &= (B \cap (A^c)^c) = B \cap A \\ B \cap A^c &= B - A \\ \Leftrightarrow \mu[B - A^c] + \mu[A^c \cap B] &= \mu[B \cap A] + \mu[B - A] \\ &= \mu[B] \end{aligned}$$

All implications are bidirectional.

(c): Suppose $\{A_k\}_{k=1}^{\infty}$ is a sequence of μ -measurable and $A_i \subseteq A_{i+1}$. Base case for $\mu[A_1]$ is trivial. Assume for the sake of induction that:

$$\sum_{k=1}^{n-1} \mu[A_{k+1} - A_k] + \mu[A_1] = \mu[A_n].$$

Now by μ -measurability of A_n , we know

$$\begin{aligned} \mu[A_{n+1}] &= \mu[A_{n+1} - A_n] + \mu[A_{n+1} \cap A_n] \\ &= \mu[A_{n+1} - A_n] + \mu[A_n] \quad (A_n \subseteq A_{n+1}) \\ &= \sum_{k=1}^n \mu[A_{k+1} - A_k] + \mu[A_1] \quad (\text{inductive hyp.}) \end{aligned}$$

(d): Suppose

$$A_i \supseteq A_{i+1},$$

$\mu[A_1] < \infty$, and all A_i are μ -measurable. A_i^c is measurable.

Problem 1.2*

Suppose μ is a measure on $\mathbb{Z}^+ := \mathbb{Z} \cap [0, +\infty)$ such that every subset of $\mathbb{Z}^+$ is μ -measurable. Show that there exists a sequence $(\omega_k)_{k=0}^\infty \subseteq [0, +\infty]$ such that for every $E \subseteq \mathbb{Z}^+$ we have

$$\mu(E) = \sum_{k \in E} \omega_k.$$

Solution. Let

$$\omega_k := \mu[\{k\}].$$

For finite sets $E \subset \mathbb{Z}^+$, showing the claim follows from an inductive argument. The case for singletons is trivial. Suppose for the sake of induction that it is true for all $E \subset \mathbb{Z}^+$ such that $|E| = k$. Then let $F = \{x_1, \dots, x_{k+1}\} \subset \mathbb{Z}^+$.

$$\begin{aligned} \mu[F] &= \mu[F - \{x_{k+1}\}] + \mu[\{x_{k+1}\}] \quad (F \text{ is measurable}) \\ &= \sum_{i=1}^k \omega_{x_i} + \omega_{x_{k+1}}. \end{aligned}$$

Now for $E \subseteq \mathbb{Z}^+$ infinite we will do the following argument. E countable, thus let $E = \{x_i\}_{i=1}^\infty$ be an enumeration. Let $A_n := \{x_1, \dots, x_n\}$. Note that $A_n \subseteq A_{n+1}$.

$$\begin{aligned} \mu[E] &= \lim_{n \rightarrow \infty} \mu[A_n] \\ &= \lim_{n \rightarrow \infty} \left(\mu[A_1] + \sum_{k=1}^{n-1} \mu[A_{k+1} - A_k] \right) \\ &= \lim_{n \rightarrow \infty} \left(\omega_{x_1} + \sum_{k=1}^{n-1} \omega_{x_k} \right) \\ &= \sum_{k=1}^\infty \omega_{x_k} = \sum_{e \in E} \omega_e \end{aligned}$$

Problem 1.3*

Suppose μ is an outer measure on X such that $\mu(X) < \infty$ and suppose $\mathcal{S}$ is the set of μ -measurable subsets. Suppose $\mathcal{A} \subseteq \mathcal{S}$ is such that for every $A \in \mathcal{A}$, $\mu(A) > 0$ and for every $A, B \in \mathcal{A}$ such that $A \neq B$ then $A \cap B = \emptyset$. Show that $\mathcal{A}$ is countable.

Solution. Let

$$I_n := \left\{ A \in \mathcal{A} : \mu[A] \geq \frac{1}{n} \right\}.$$

Note that

$$\mathcal{A} = \bigcup_{n \in \mathbb{N}} I_n.$$

Assume the contrary that I_n is not finite. Let $J \subseteq I_n$ be a countably infinite subset,

$$\mu[X] \geq \mu\left[\bigcup_{A \in I_n} A\right] \geq \mu\left[\bigcup_{A \in J} A\right] = \sum_{A \in J} \mu[A] \geq \sum_{A \in J} \frac{1}{n} = \infty.$$

Thus a contradiction occurs. Hence $|I_n|$ is finite. Hence $\mathcal{A}$ is a countable union of finite sets hence must be finite.

Problem 1.4*

Let μ be an outer measure on a set X and let S be the collection of μ -measurable subsets on X . Show that we cannot have

$$\{\mu(E) : E \in S\} = [0, 1).$$

Solution. Note that $X \in S$. Thus $\exists \alpha \in [0, 1)$ where $\mu[X] = \alpha$. But since we have a bijection between $\{\mu(E) : E \in S\} = [0, 1)$ there exists a $E \in S$ such that $\mu(S) = \frac{\alpha+1}{2}$. But $E \subseteq X$ and by monotonicity of outer measure, $\frac{\alpha+1}{2} = \mu(E) \leq \mu(X) = \alpha$ this is a contradiction.

Problem 1.5*

Let μ be a measure on a set X such that $\mu[X] = 1$ and let $\mathcal{M}$ be the set of μ -measurable sets. Suppose that $\mu[M] > 0$ for each $\emptyset \neq M \in \mathcal{M}$ and let

$$\alpha(x) = \inf_{M \in \mathcal{M}} \{\mu[M] : x \in M\} \quad (x \in X).$$

- (a) Show that there exists a set $A_x \in \mathcal{M}$ such that $x \in A_x$ and $\mu[A_x] = \alpha(x)$.
- (b) Show that the sets $\{A_x\}$ are either disjoint or identical.

Solution. (a): We know that following that: $\forall n \in \mathbb{N}$ there exists a $E_n \in \mathcal{M}$ such that

$$\mu(E_n) \leq \alpha(x) + \frac{1}{n}$$

and $x \in E_n$.

$$\bigcap_{n \in \mathbb{N}} E_n \ni \{x\}.$$

Consider

$$F_k := \bigcap_{n=1}^k E_k.$$

Note that $F_1 \supseteq F_2 \supseteq \dots$, and $\{x\} \in F_k \Rightarrow \alpha(x) \leq \mu(F_n) \leq \mu(E_n) = \alpha(x) + \frac{1}{n}$. By problem 1, we know that

$$\lim_{n \rightarrow \infty} \mu[F_n] = \mu\left[\bigcap_{n \in \mathbb{N}} F_n\right].$$

By the squeeze theorem we know, that

$$\alpha(x) \leq \mu \left[\bigcap_{n \in \mathbb{N}} F_n \right] \leq \lim_{n \rightarrow \infty} \left(\alpha(x) + \frac{1}{n} \right) = \alpha(x) \Rightarrow \mu \left[\bigcap_{n \in \mathbb{N}} F_n \right] = \alpha(x).$$

Furthermore, we know that

$$\bigcap_{n \in \mathbb{N}} F_n = \bigcap_{n \in \mathbb{N}} E_n \text{ is measurable..}$$

Thus

$$A_x = \bigcap_{n \in \mathbb{N}} F_n$$

(b): Suppose $x, y \in X$ where $A_x \neq A_y$. Assume the contrary, that $A_x \cap A_y \neq \emptyset$.

Case 1:

Problem 1.6

Let μ be a measure on X such that $\mu[X] < \infty$. Show that if $\{x\}$ is μ -measurable for every $x \in X$ then there are at most countably many $x \in X$ such that $\mu[\{x\}] > 0$.

Solution.

Problem 1.7*

Show that $S = \{\cup_{n \in K} (n, n+1] : K \subseteq \mathbb{Z}\} \cup \{\emptyset\}$ is a σ -algebra on $\mathbb{R}$.

Solution.

Problem 1.8*

Prove that $\sigma(C_1) = \sigma(C_2) = \sigma(C_3) = \mathcal{B}$, where $\mathcal{B}$ is the Borel σ -algebra on $\mathbb{R}$. Here,

$$C_1 = \{(r, s] : r, s \in \mathbb{Q}\}, \quad C_2 = \{(r, n] : r \in \mathbb{Q}, n \in \mathbb{Z}\}, \quad C_3 = \{[r, +\infty) : r \in \mathbb{Q}\}.$$

Solution.

Problem 1.9*

Prove that if $B \subseteq \mathbb{R}^d$ is a Borel set, $t \in \mathbb{R}$ and $c \in \mathbb{R}^d$ then $tB = \{tx : x \in B\}$ and $c + B = \{c + x : x \in B\}$ are Borel sets, where $\mathcal{B}$ is the Borel σ -algebra on $\mathbb{R}$.

Solution.

Problem 1.10

Let (X, μ) be an outer measure space and $(A_n)_n$ be a sequence of μ -measurable subsets of X . Show that if $\sum_{n=1}^{\infty} \mu[A_n] < \infty$ then $\mu[A] = 0$ if we set

$$A = \cap_{m=1}^{\infty} \cup_{n=m}^{\infty} A_n.$$

Solution.